

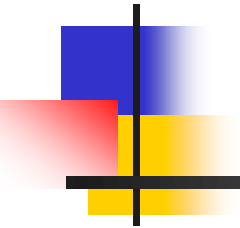


Indian Institute of Information Technology, Surat

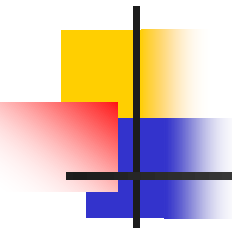
भारतीय सूचना प्रौद्योगिकी संस्थान, सूरत
(Institute of National Importance under Act of Parliament)

EC 503: Image Processing and Computer Vision

By:
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ECE Department
IIIT Surat

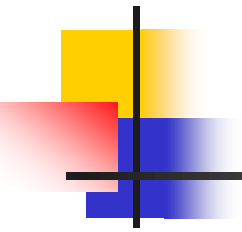


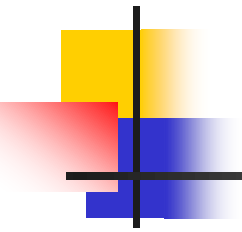
Morphological Image Processing

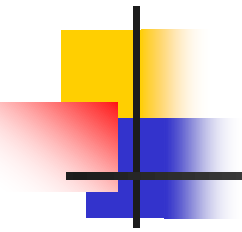


Introduction

- **Morphology:** a branch of biology that deals with the form and structure of animals and plants
- Morphological image processing is used to extract image components for representation and description of region shape, such as boundaries, skeletons, and the convex hull
- Morphological image processing is a set of techniques that use shapes to analyze and modify the structure and shape of objects in an image. It's based on the mathematical theory of morphology, which studies the properties of patterns and shapes.

- 
- The language of mathematical morphology is set theory. As such, morphology offers a unified and powerful approach to numerous image processing problems. Sets in mathematical morphology represent objects in an image.
 - For example, the set of all white pixels in a binary image is a complete morphological description of the image.

- 
- If S is the set of pixels (2-D points) representing an object in an image, then S' is simply the set of points in whose coordinates have been replaced by $(-x, -y)$ -- Reflection
 - If S is the set of pixels representing an object in an image, then S' is the set of points in whose coordinates have been replaced by $(x + z_1, y + z_2)$ -- Translation



Preliminaries (1)

■ Reflection

The reflection of a set B , denoted $\hat{B}$, is defined as

$$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$$

■ Translation

The translation of a set B by point $z = (z_1, z_2)$, denoted $(B)_z$, is defined as

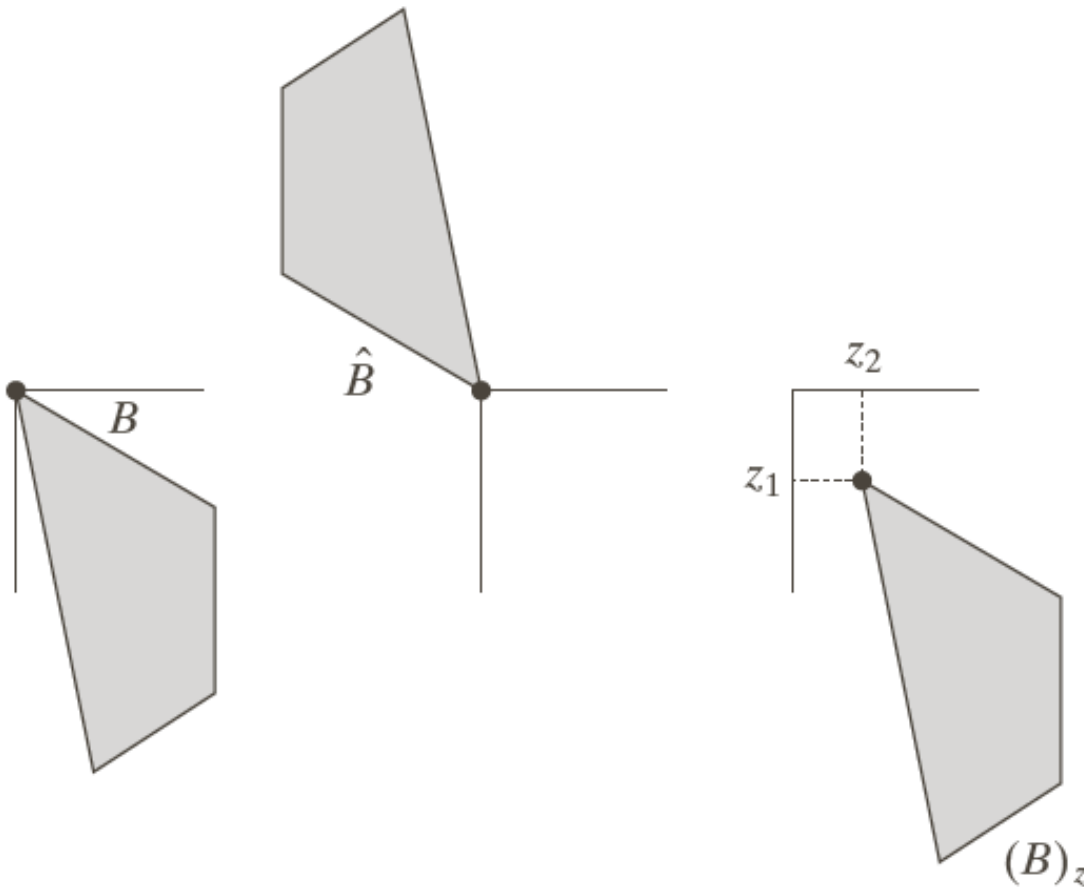
$$(B)_z = \{c \mid c = b + z, \text{ for } b \in B\}$$

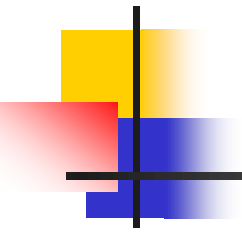
Example: Reflection and Translation

a b c

FIGURE 9.1

(a) A set, (b) its reflection, and (c) its translation by z .





Preliminaries (2)

- **Structure elements (SE)**

Set reflection and translation are employed extensively in morphology to formulate operations based on so-called structuring elements (SEs): small sets or subimages used to probe an image under study for properties of interest.

Examples: Structuring Elements (1)

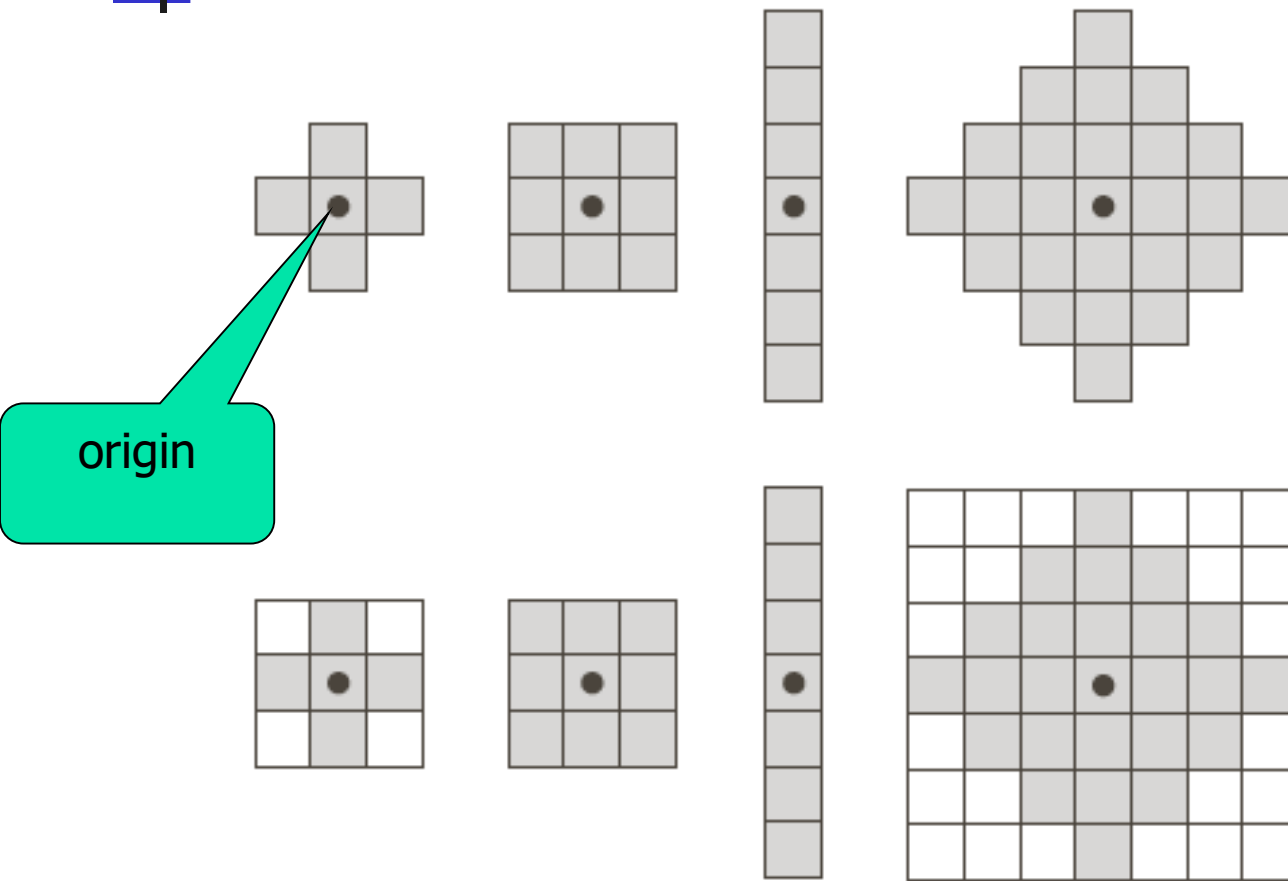
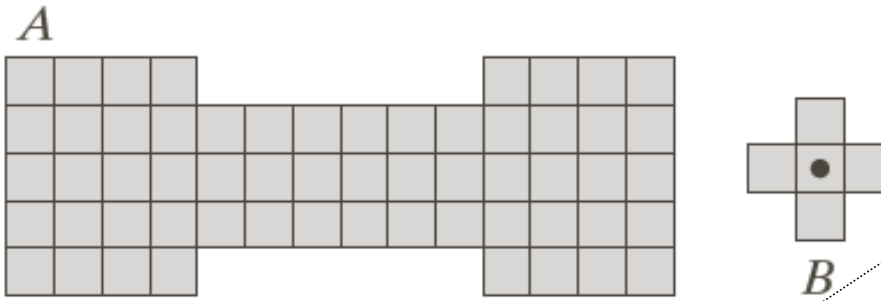


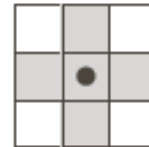
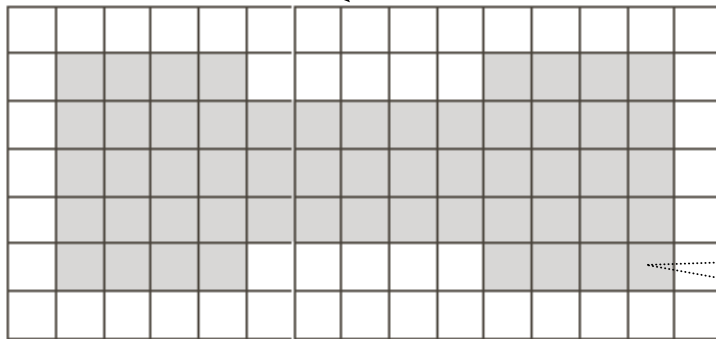
FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.

Examples: Structuring Elements (2)

Accommodate the entire structuring element when its origin is on the border of the original set A



Origin of B visits every element of A



At each location of the origin of B, if B is completely contained in A, then the location is a member of the new set, otherwise it is not a member of the new set.

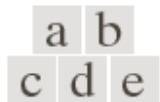
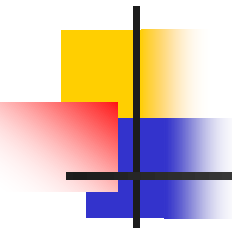


FIGURE 9.3 (a) A set (each shaded square is a member of the set). (b) A structuring element. (c) The set padded with background elements to form a rectangular array and provide a background border. (d) Structuring element as a rectangular array. (e) Set processed by the structuring element.



Erosion

With A and B as sets in Z^2 , the erosion of A by B , denoted $A \ominus B$, defined

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

- In words, this equation indicates that the erosion of A by B is the set of all points z such that B , translated by z , is contained in A . In the following discussion, set B is assumed to be a structuring element.

Because the statement that B has to be contained in A is equivalent to B not sharing any common elements with the background, we can express erosion in the following equivalent form:

$$A \ominus B = \{z \mid (B)_z \cap A^c = \emptyset\}$$

Where A^c is the complement of A and $\emptyset$ is the empty set.

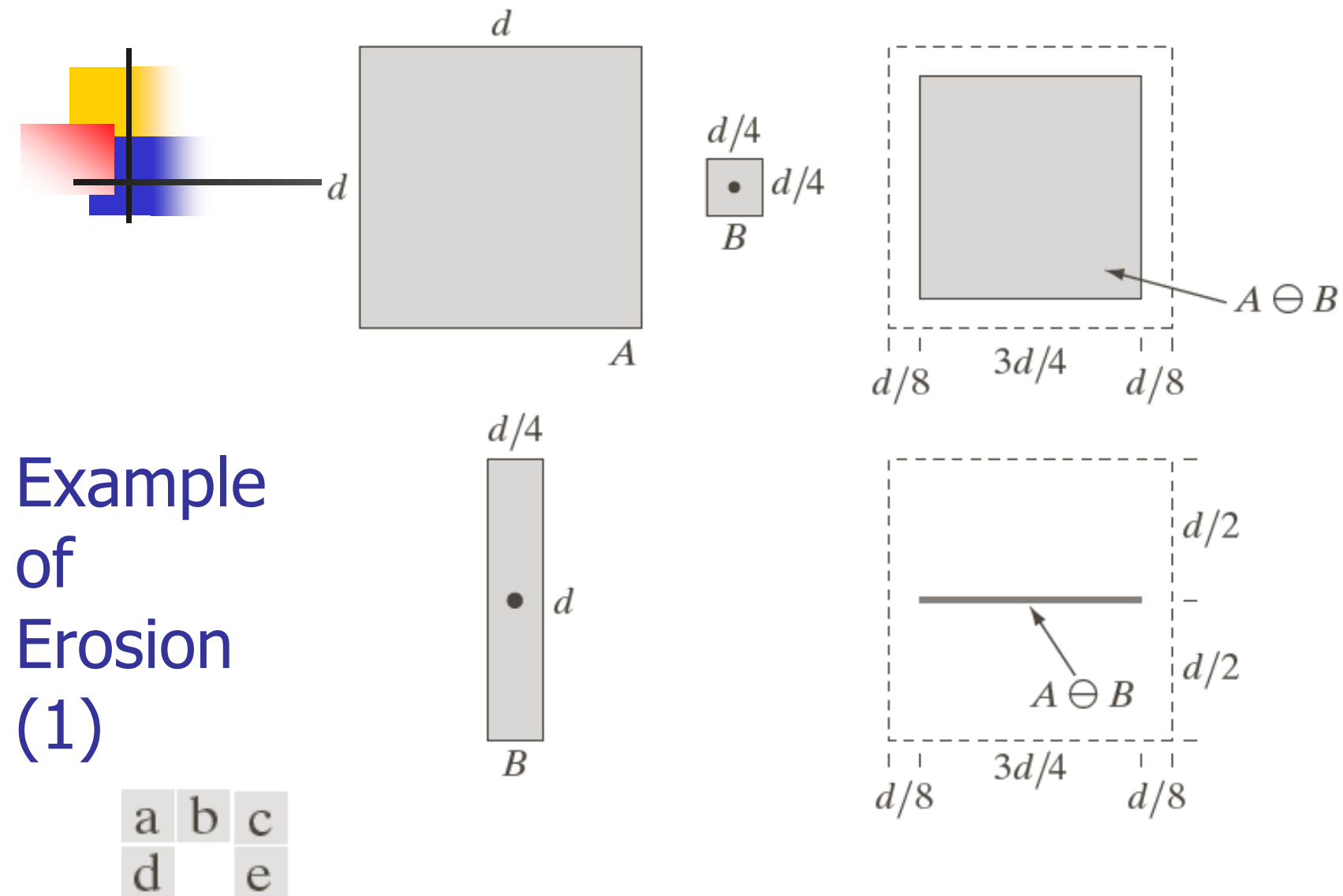
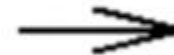
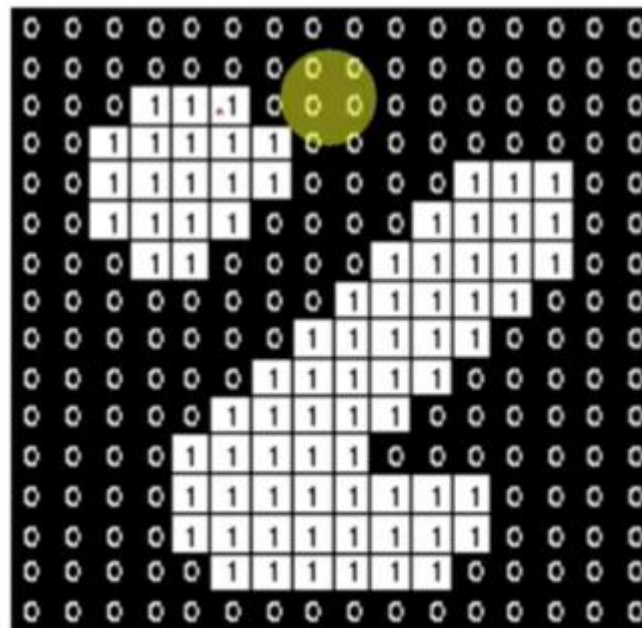
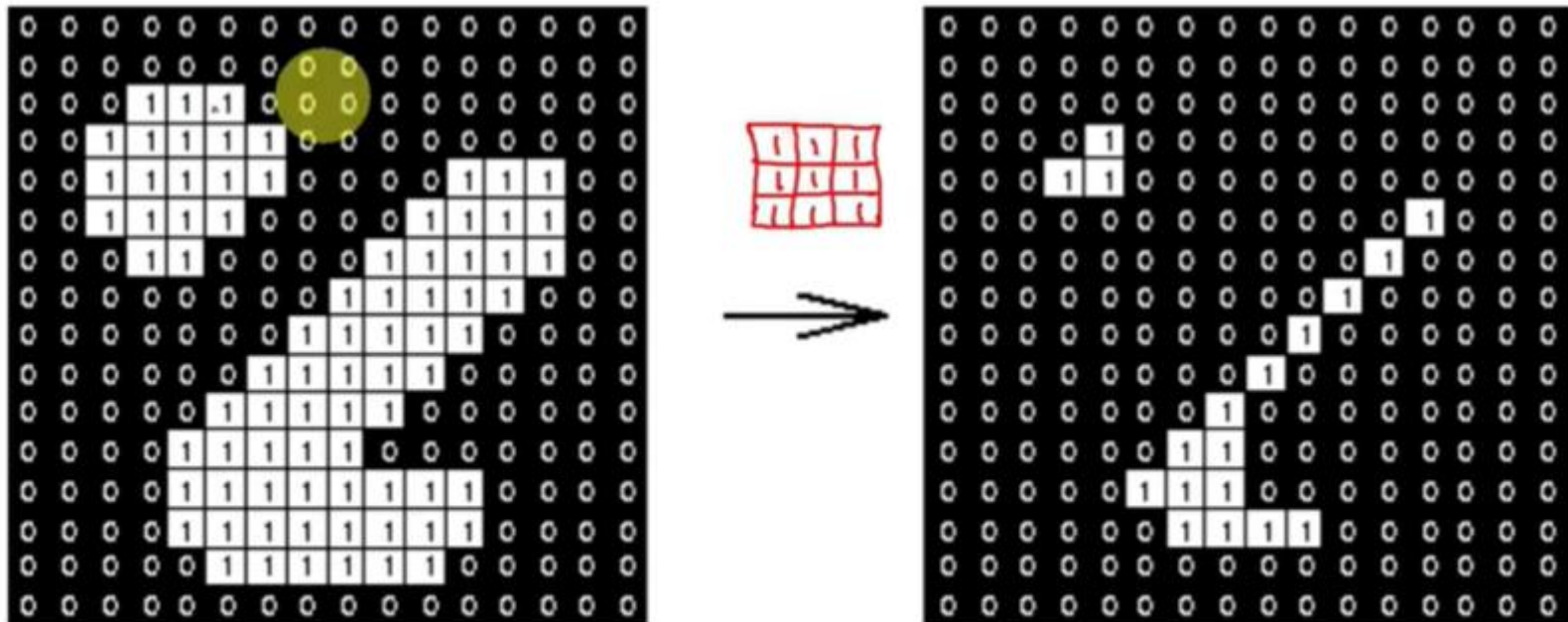


FIGURE 9.4 (a) Set A . (b) Square structuring element, B . (c) Erosion of A by B , shown shaded. (d) Elongated structuring element. (e) Erosion of A by B using this element. The dotted border in (c) and (e) is the boundary of set A , shown only for reference.

Example



Example



Erosion: a 3×3 square structuring element

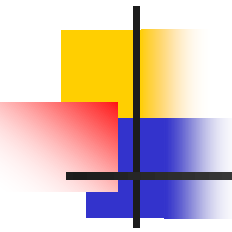


Example: Use the following structuring element to erode the binary image below.

1
1
1
Structuring
element

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

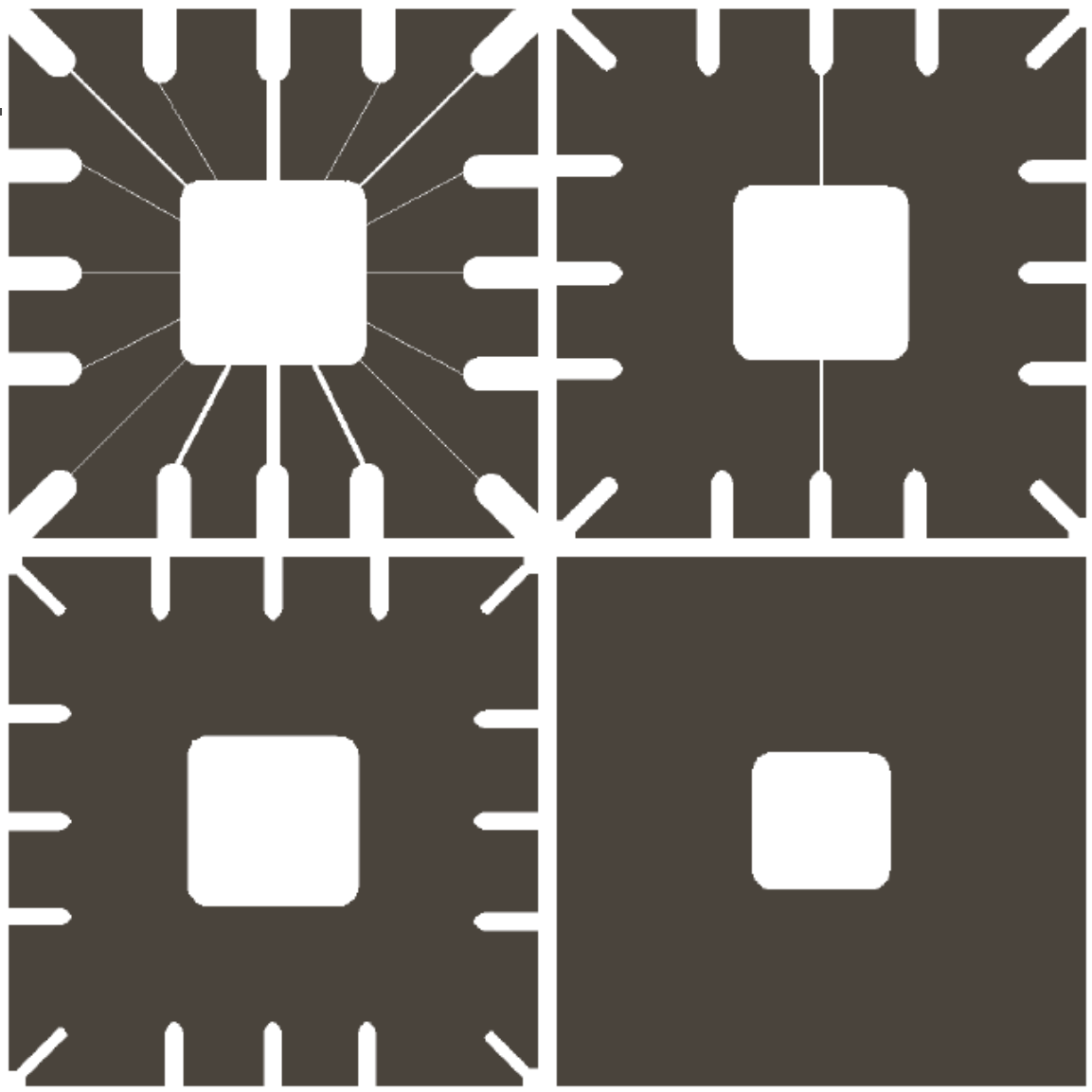
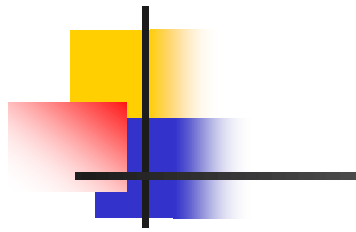
Binary image



Solution

$$A \ominus B =$$

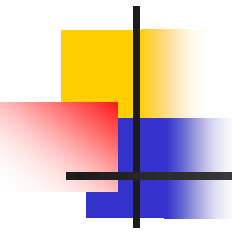
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0



a	b
c	d

FIGURE 9.5 Using erosion to remove image components. (a) A 486×486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11×11 , 15×15 , and 45×45 , respectively. The elements of the SEs were all 1s.

Example of Erosion (2)



Dilation

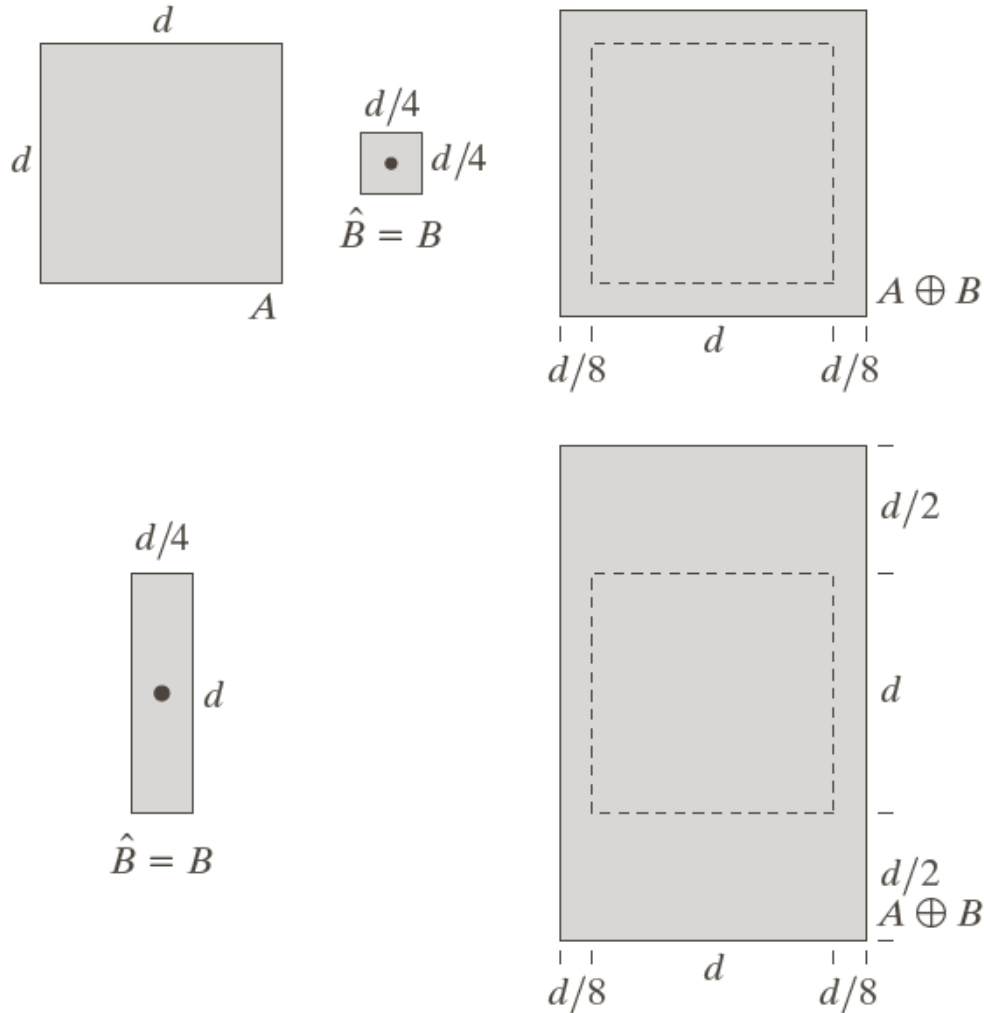
With A and B as sets in Z^2 , the dilation of A by B , denoted $A \oplus B$, is defined as

$$A \oplus B = \{z | (\hat{B})_z \cap A \neq \emptyset\}$$

The set of all displacements z , the translated B and A overlap by at least one element.

$$A \oplus B = \{z | [(\hat{B})_z \cap A] \subseteq A\}$$

Examples of Dilation (1)

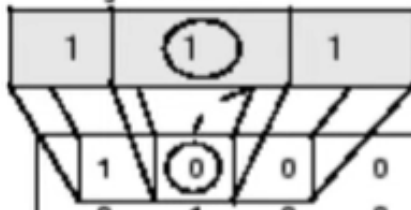


a	b	c
d		e

FIGURE 9.6

(a) Set A .
 (b) Square structuring element (the dot denotes the origin).
 (c) Dilation of A by B , shown shaded.
 (d) Elongated structuring element.
 (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A , shown only for reference

Structuring Element



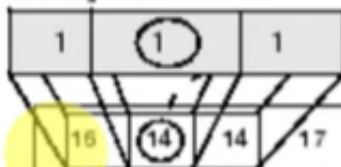
1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

Input Image

1	1
1	
0	
0	
0	

Output Image

Structuring Element



16	14	14	17	19	15	21
53	57	61	62	64	60	68
126	128	124	122	125	125	127
132	130	133	132	131	132	130
140	138	137	143	138	137	134
143	141	138	142	140	134	144
138	142	137	139	138	132	136

Input Image

16	16
57	
128	
132	
140	
143	
142	

Output Image

Examples of Dilation (2)

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



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0	1	0
1	1	1
0	1	0

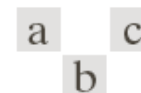
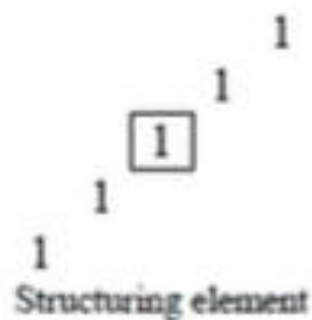


FIGURE 9.7

(a) Sample text of poor resolution with broken characters (see magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

Example: Use the following structuring element to dilate the binary image below.



0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

Binary image

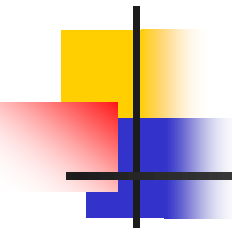
We find the reflection of B:

$$B = \begin{matrix} & & & & 1 \\ & & & & & 1 \\ & & & & & & 1 \\ & & & & & & & 1 \\ & & & & & & & & 1 \\ & & & & & & & & & 1 \\ & & & & & & & & & & 1 \\ & & & & & & & & & & & 1 \\ & & & & & & & & & & & & 1 \\ & & & & & & & & & & & & & 1 \\ & & & & & & & & & & & & & & 1 \\ & & & & & & & & & & & & & & & 1 \end{matrix}$$

In this case $\hat{B} = B$

$$A \oplus B =$$

0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1	1	0
0	0	0	0	1	1	1	1	1	1	1	1	0
0	0	0	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	0	0	0
0	1	1	1	1	1	1	1	1	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0	0
0	1	1	1	1	1	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0

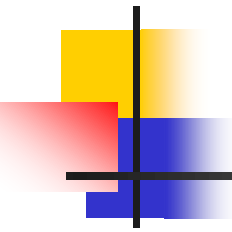


Duality

- Erosion and dilation are duals of each other with respect to set complementation and reflection

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

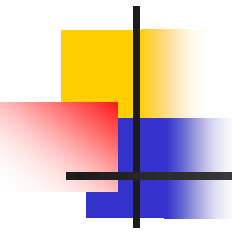
$$(A \oplus B)^c = A^c \ominus \hat{B}$$



Duality

- Erosion and dilation are duals of each other with respect to set complementation and reflection

$$\begin{aligned}(A \ominus B)^c &= \left\{ z \mid (B)_z \subseteq A \right\}^c \\&= \left\{ z \mid (B)_z \cap A^c = \emptyset \right\}^c \\&= \left\{ z \mid (B)_z \cap A^c \neq \emptyset \right\} \\&= A^c \oplus \hat{B}\end{aligned}$$



Duality

- Erosion and dilation are duals of each other with respect to set complementation and reflection

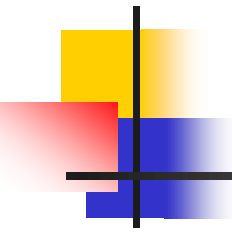
$$\begin{aligned}(A \oplus B)^c &= \left\{ z \mid \left(\hat{B} \right)_z \cap A \neq \emptyset \right\}^c \\ &= \left\{ z \mid \left(\hat{B} \right)_z \cap A^c = \emptyset \right\} \\ &= A^c \ominus \hat{B}\end{aligned}$$

Duality



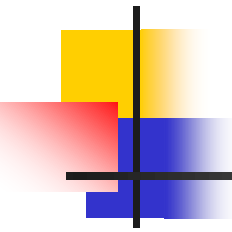
a b c

FIGURE 9.7 (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side. (b) Erosion of (a) with a square structuring element of 1's, 13 pixels on the side. (c) Dilation of (b) with the same structuring element.



Opening and Closing

- Opening generally smoothes the contour of an object, breaks narrow isthmuses, and eliminates thin protrusions
- Closing tends to smooth sections of contours but it generates fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps in the contour



Opening and Closing

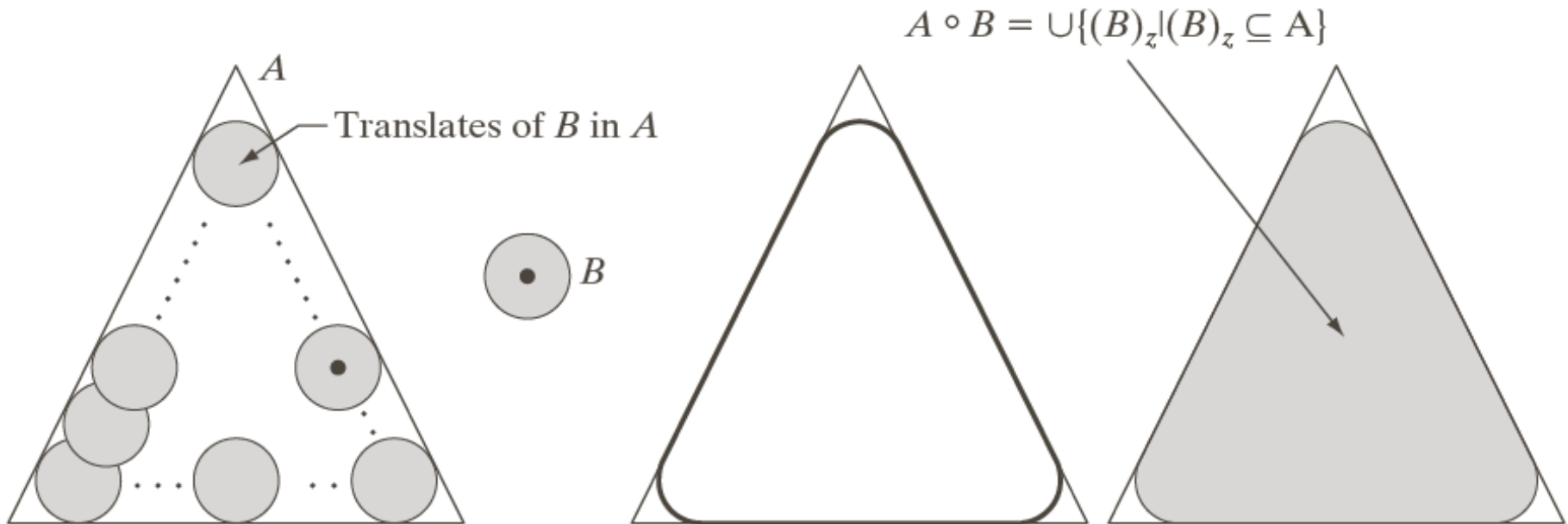
The opening of set A by structuring element B , denoted $A \circ B$, is defined as

$$A \circ B = (A \ominus B) \oplus B$$

The closing of set A by structuring element B , denoted $A \bullet B$, is defined as

$$A \bullet B = (A \oplus B) \ominus B$$

Example: Opening



a b c d

FIGURE 9.8 (a) Structuring element B “rolling” along the inner boundary of A (the dot indicates the origin of B). (b) Structuring element. (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded). We did not shade A in (a) for clarity.

Example: Closing

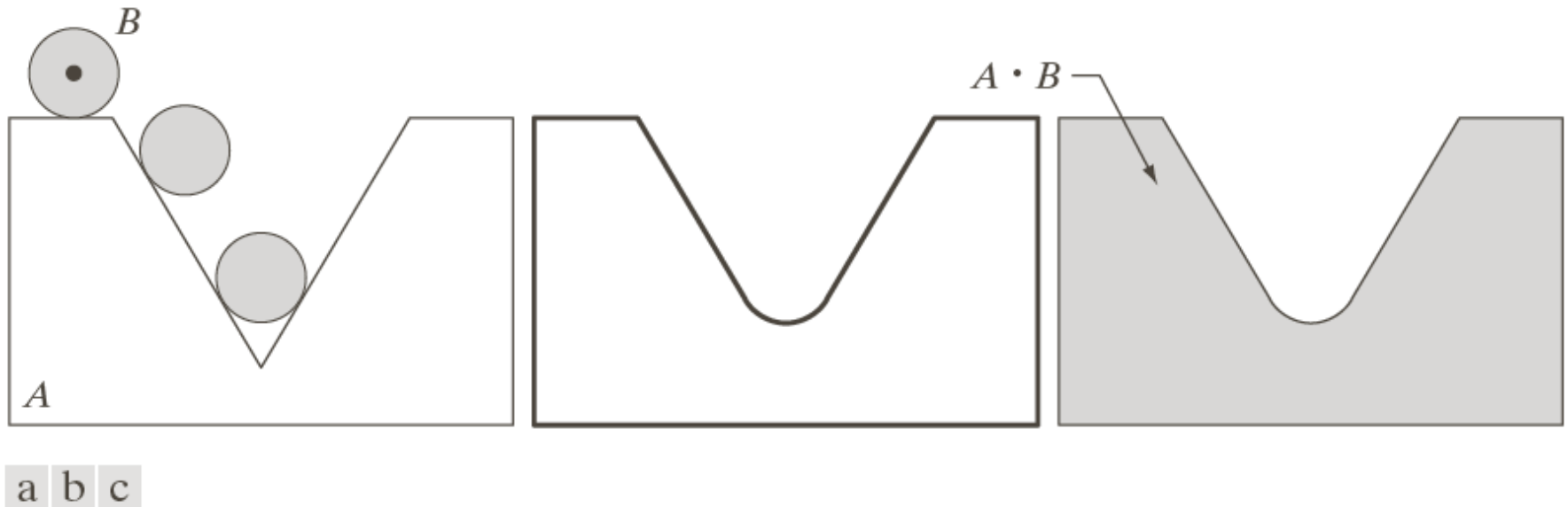
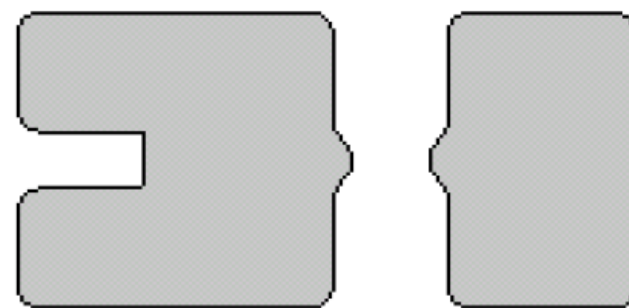
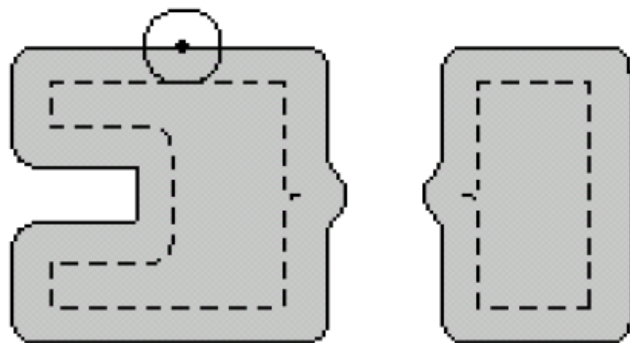
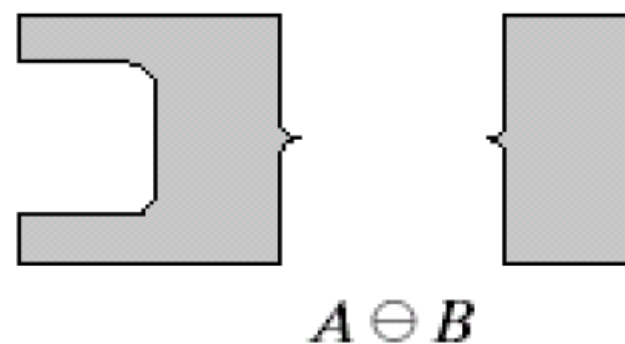
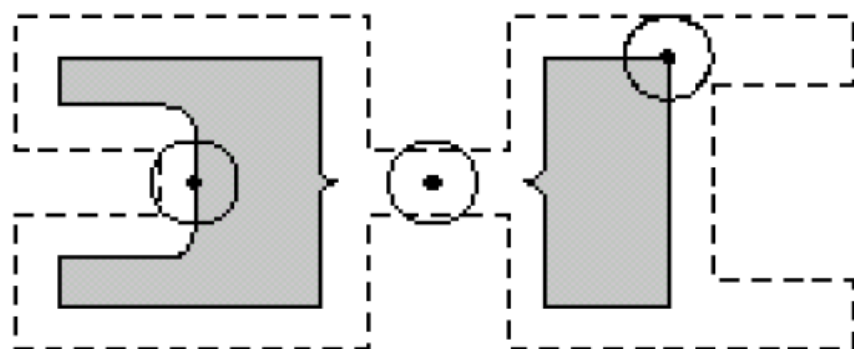
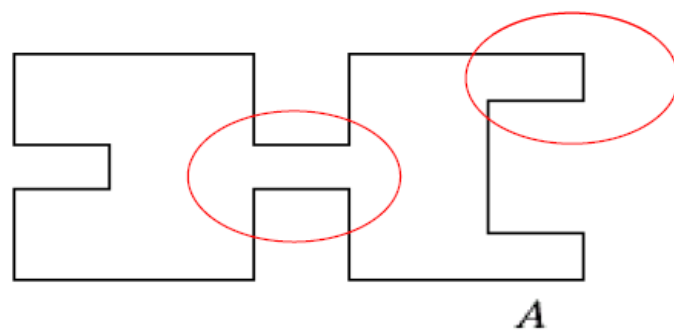


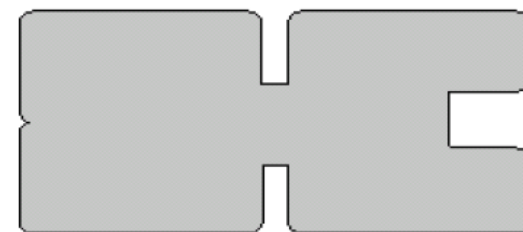
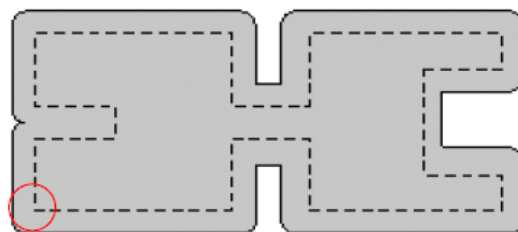
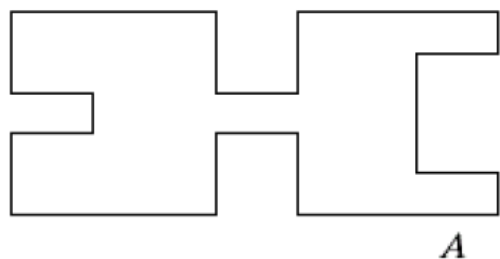
FIGURE 9.9 (a) Structuring element B “rolling” on the outer boundary of set A . (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.

Opening (cont.)

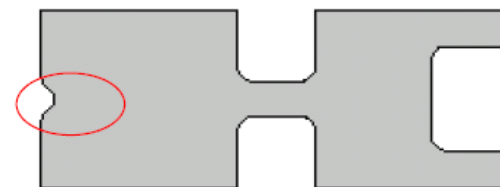
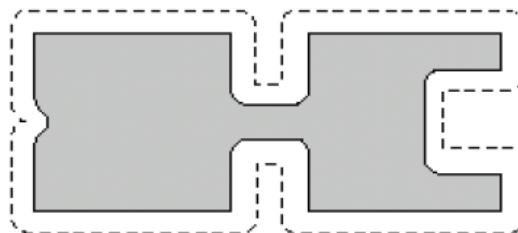


$$A \circ B = (A \ominus B) \oplus B$$

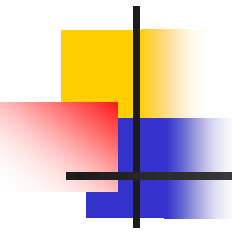
Closing (cont.)



$$A \oplus B$$



$$A \cdot B = (A \oplus B) \ominus B$$

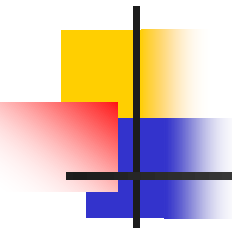


Duality of Opening and Closing

- Opening and closing are duals of each other with respect to set complementation and reflection

$$(A \bullet B)^c = (A^c \circ \hat{B})$$

$$(A \circ B)^c = (A^c \bullet \hat{B})$$



The Properties of Opening and Closing

- Properties of Opening

- (a) $A \circ B$ is a subset (subimage) of A .
- (b) If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$.
- (c) $(A \circ B) \circ B = A \circ B$.

- Properties of Closing

- (a) A is a subset (subimage) of $A \bullet B$.
- (b) If C is a subset of D , then $C \bullet B$ is a subset of $D \bullet B$.
- (c) $(A \bullet B) \bullet B = A \bullet B$.

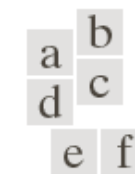
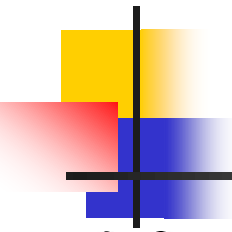


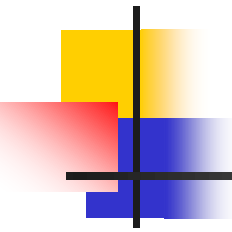
FIGURE 9.11

(a) Noisy image.
 (b) Structuring element.
 (c) Eroded image.
 (d) Opening of A.
 (e) Dilation of the opening.
 (f) Closing of the opening.
 (Original image courtesy of the National Institute of Standards and Technology.)



The Hit-or-Miss transformation

- A basic morphological tool for shape detection.
- Let the origin of each shape be located at its center of gravity.
- If we want to find the location of a shape , say – X , at (larger) image, say – A :
 - ▣ Let X be enclosed by a small window, say – W .
 - ▣ The **local background** of X with respect to W is defined as the *set difference* ($W - X$).
 - ▣ Apply *erosion* operator of A by X , will get us the set of locations of the origin of X , such that X is completely contained in A .
 - ▣ It may be also view geometrically as the set of all locations of the origin of X at which X found a match (hit) in A .



The Hit-or-Miss transformation

Cont.

- Apply erosion operator on the complement of A by the local background set $(W - X)$.
- Notice, that the set of locations for which X **exactly** fits inside A is the ***intersection*** of these two last operators above.

This intersection is precisely the location sought.

Formally:

If B denotes the set composed of X and it's background –

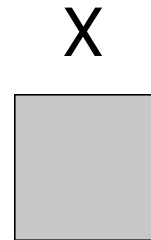
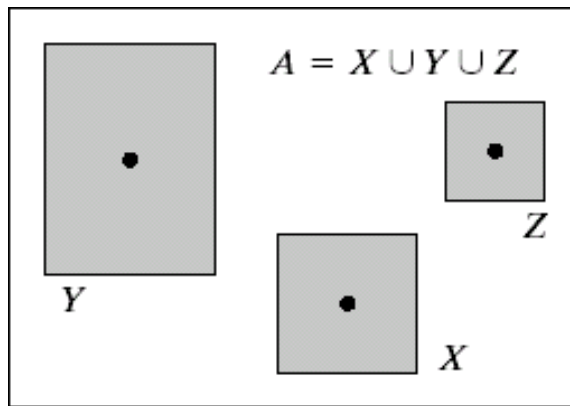
$$B = (B_1, B_2) ; B_1 = X , B_2 = (W - X).$$

The match (or set of matches) of B in A , denoted $A \circledast B$ is:

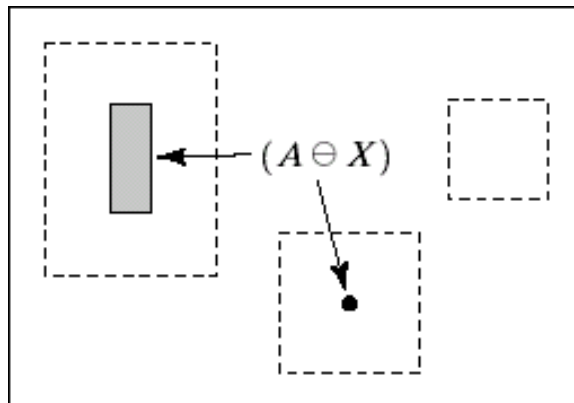
$$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

Hit-or-Miss transformation

- Find the **location of certain shape**

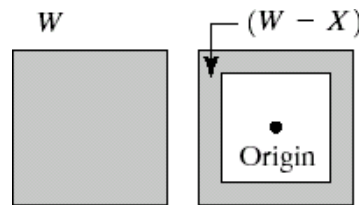
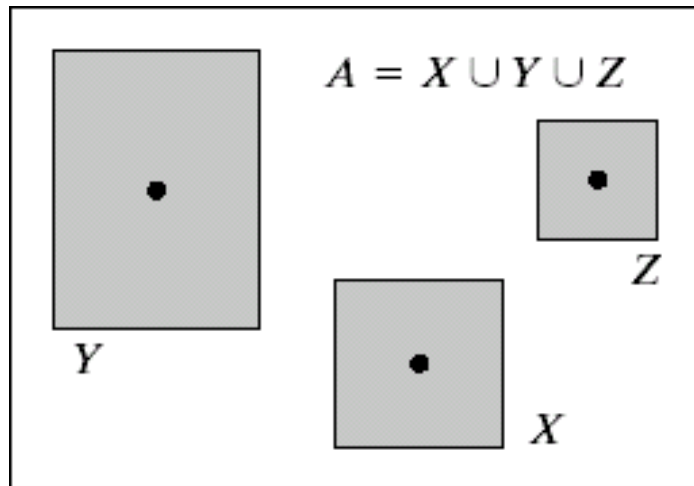


erosion

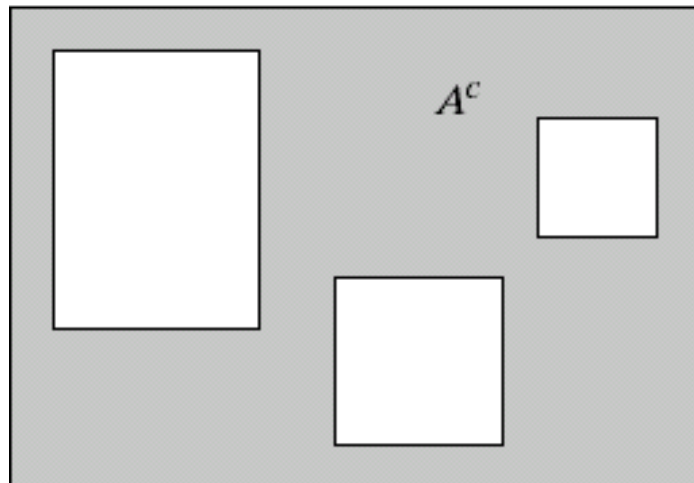


Find the set of pixels that
contain shape X

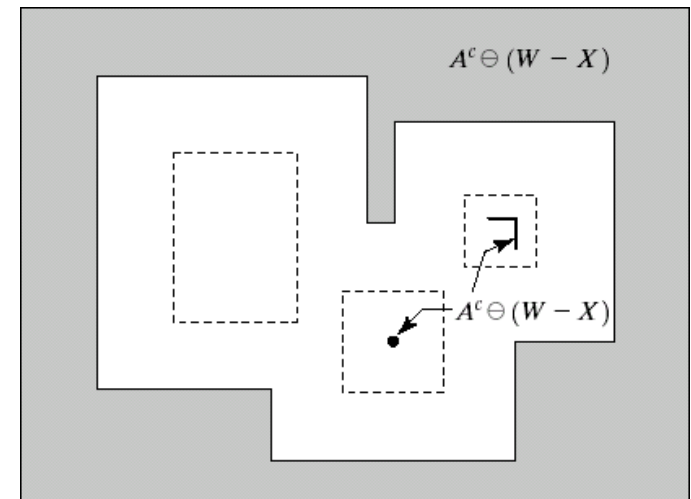
Hit-or-miss transformation



Detect object via
background

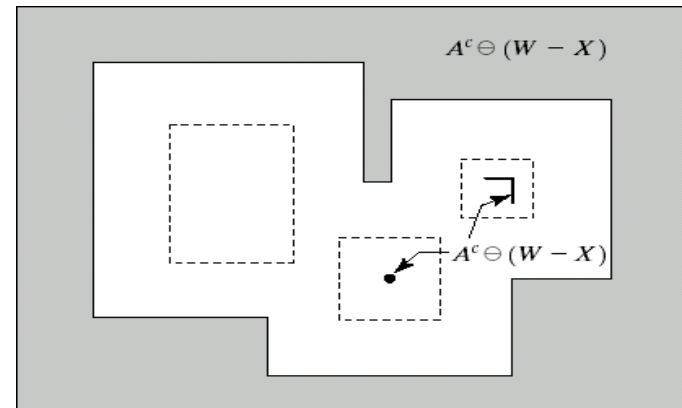
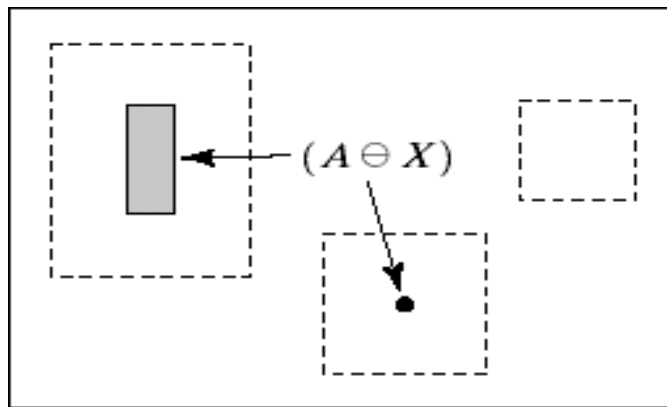


Erosion
with $(W - X)$

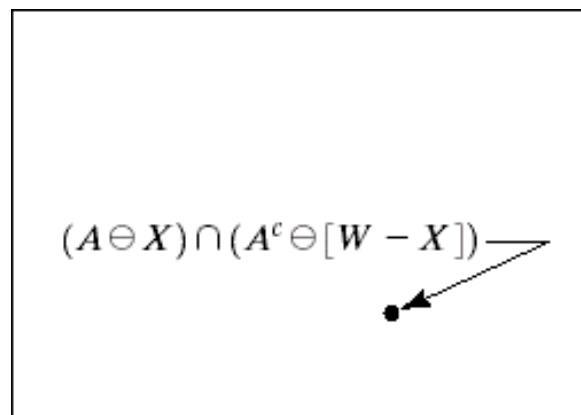


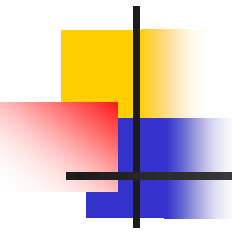
Hit-or-miss transformation

- Eliminate un-necessary parts



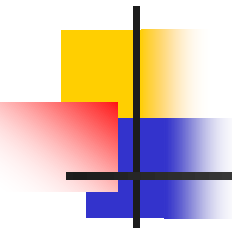
AND





The Hit-or-Miss transformation

- ▣ The reason for using these kind of structuring element – $B = (B_1, B_2)$ is based on an assumed definition that, **two or more objects are distinct only if they are disjoint (disconnected) sets.**
- ▣ In some applications , we may interested in detecting **certain patterns (combinations)** of 1's and 0's and not for detecting individual objects.
- ▣ In this case a background is not required.
and the *hit-or-miss transform* reduces to simple erosion.
- ▣ This simplified pattern detection scheme is used in some of the algorithms for – **identifying characters within a text.**



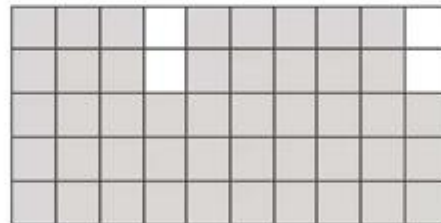
Some Basic Morphological Algorithms

- Boundary Extraction
- Region Filling
- Extraction of Connected Components
- Convex Hull
- Thinning
- Thickening
- Skeletons

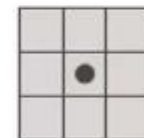
Boundary Extraction

- First, erode A by B, then make set difference between A and the erosion
- The thickness of the contour depends on the size of constructing object – B

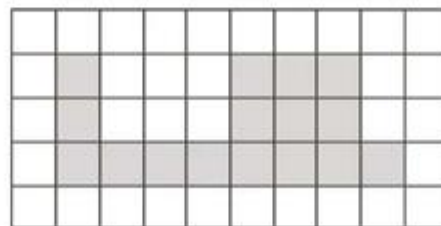
$$\beta(A) = A - (A \ominus B)$$



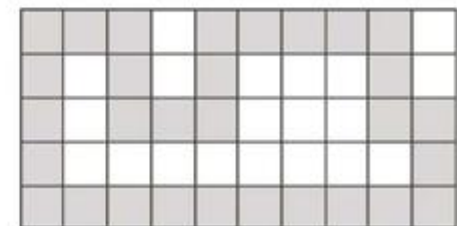
A



B



$A \ominus B$



$\beta(A)$

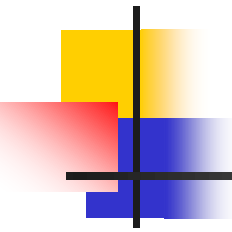
Example



a b

FIGURE 9.14

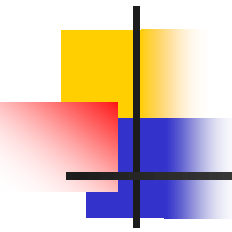
(a) A simple binary image, with 1s represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).



Region Filling/Hole Filling

- A hole may be defined as a background region surrounded by a connected border of foreground pixels.

Let A denote a set whose elements are 8-connected boundaries, each boundary enclosing a background region (i.e., a hole). Given a point in each hole, the objective is to fill all the holes with 1s.

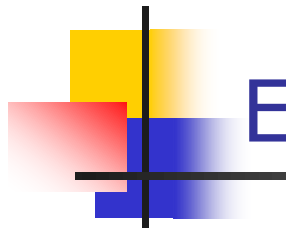


Region Filling/Hole Filling

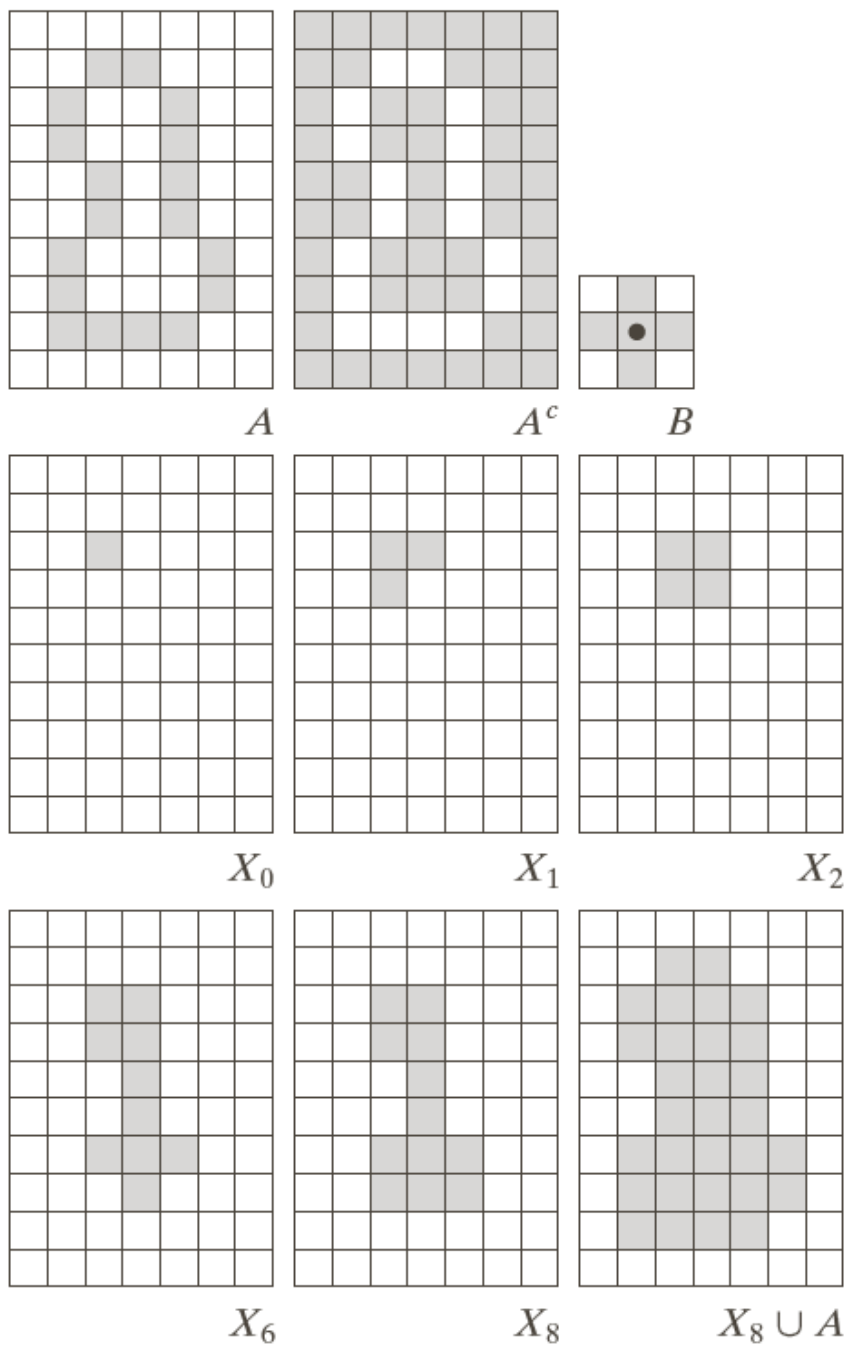
- 1. Forming an array X_0 of 0s (the same size as the array containing A), except the locations in X_0 corresponding to the given point in each hole, which we set to 1.

$$2. X_k = (X_{k-1} \oplus B) \cap A^c \quad k=1,2,3,\dots$$

Stop the iteration if $X_k = X_{k-1}$



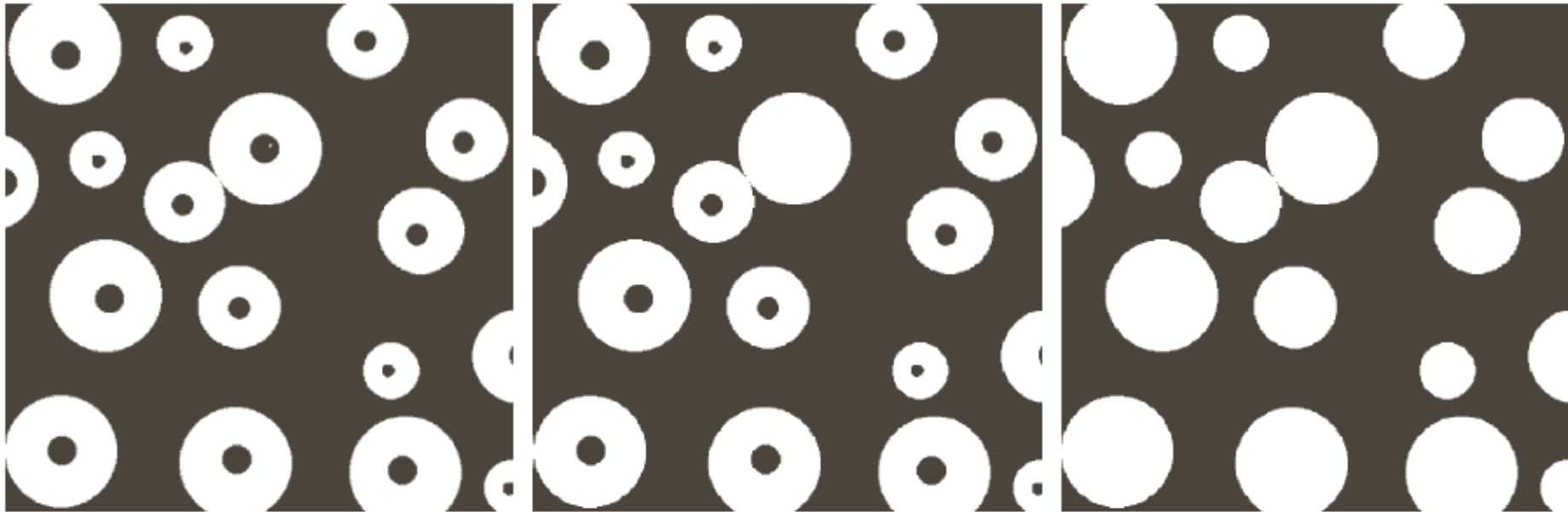
E



a	b	c
d	e	f
g	h	i

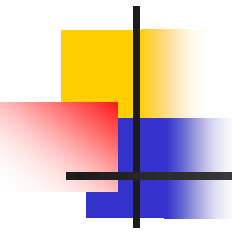
FIGURE 9.15 Hole filling. (a) Set A (shown shaded). (b) Complement of A . (c) Structuring element B . (d) Initial point inside the boundary. (e)–(h) Various steps of Eq. (9.5-2). (i) Final result [union of (a) and (h)].

Example



a b c

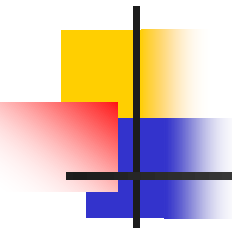
FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the hole-filling algorithm). (b) Result of filling that region. (c) Result of filling all holes.



Extraction of Connected Components

This algorithm extracts a component by selecting a point on a binary object A

Works similar to region filling, but this time we use in the conjunction the object A, instead of it's complement

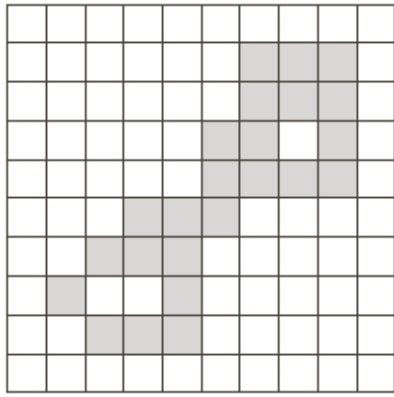
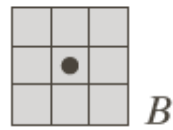
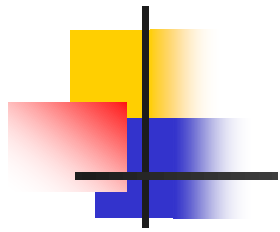


Extraction of Connected Components

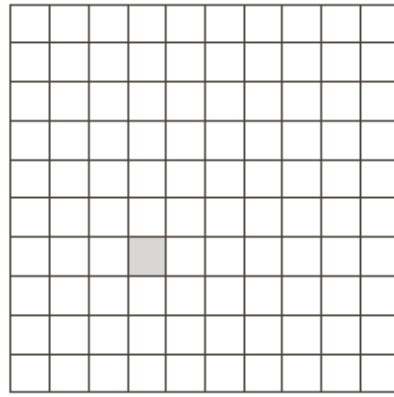
$$X_k = (X_{k-1} \oplus B) \cap A$$

B : structuring element

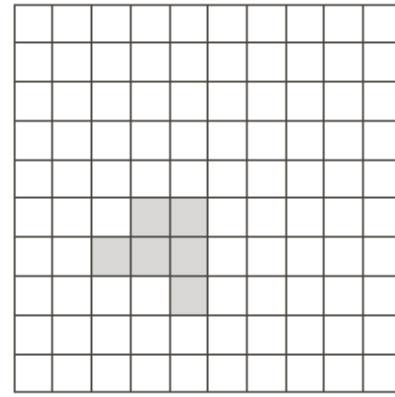
until $X_k = X_{k-1}$



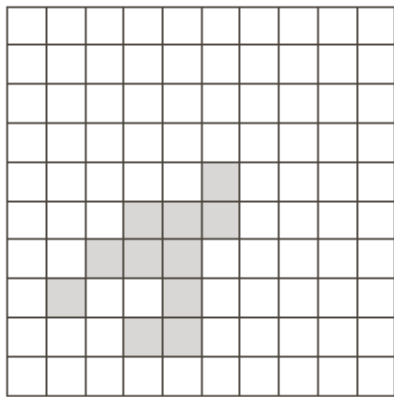
A



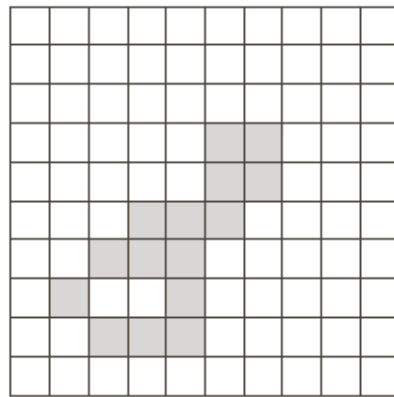
X_0



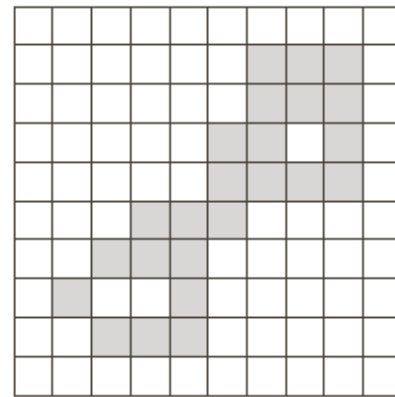
X_1



X_2



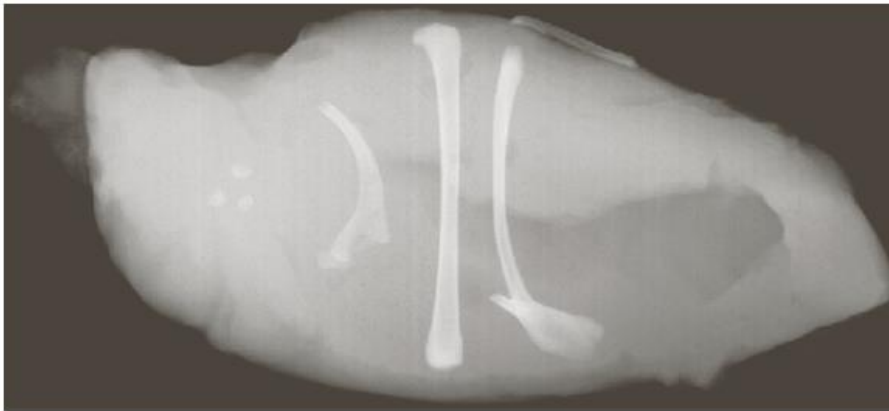
X_3



X_6



FIGURE 9.17 Extracting connected components. (a) Structuring element. (b) Array containing a set with one connected component. (c) Initial array containing a 1 in the region of the connected component. (d)–(g) Various steps in the iteration of Eq. (9.5-3).



a
b
c d

FIGURE 9.18

(a) X-ray image of chicken file with bone fragments.

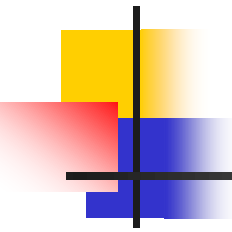
(b) Thresholded image. (c) Image eroded with a 5×5 structuring element of 1s.

(d) Number of pixels in the connected components of (c).

(Image courtesy of NTB

Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)

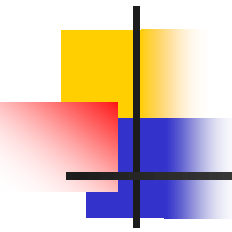
Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85



Convex Hull

A set A is said to be ***convex*** if the straight line segment joining any two points in A lies entirely within A .

The ***convex hull*** H or of an arbitrary set S is the smallest convex set containing S .



Convex Hull

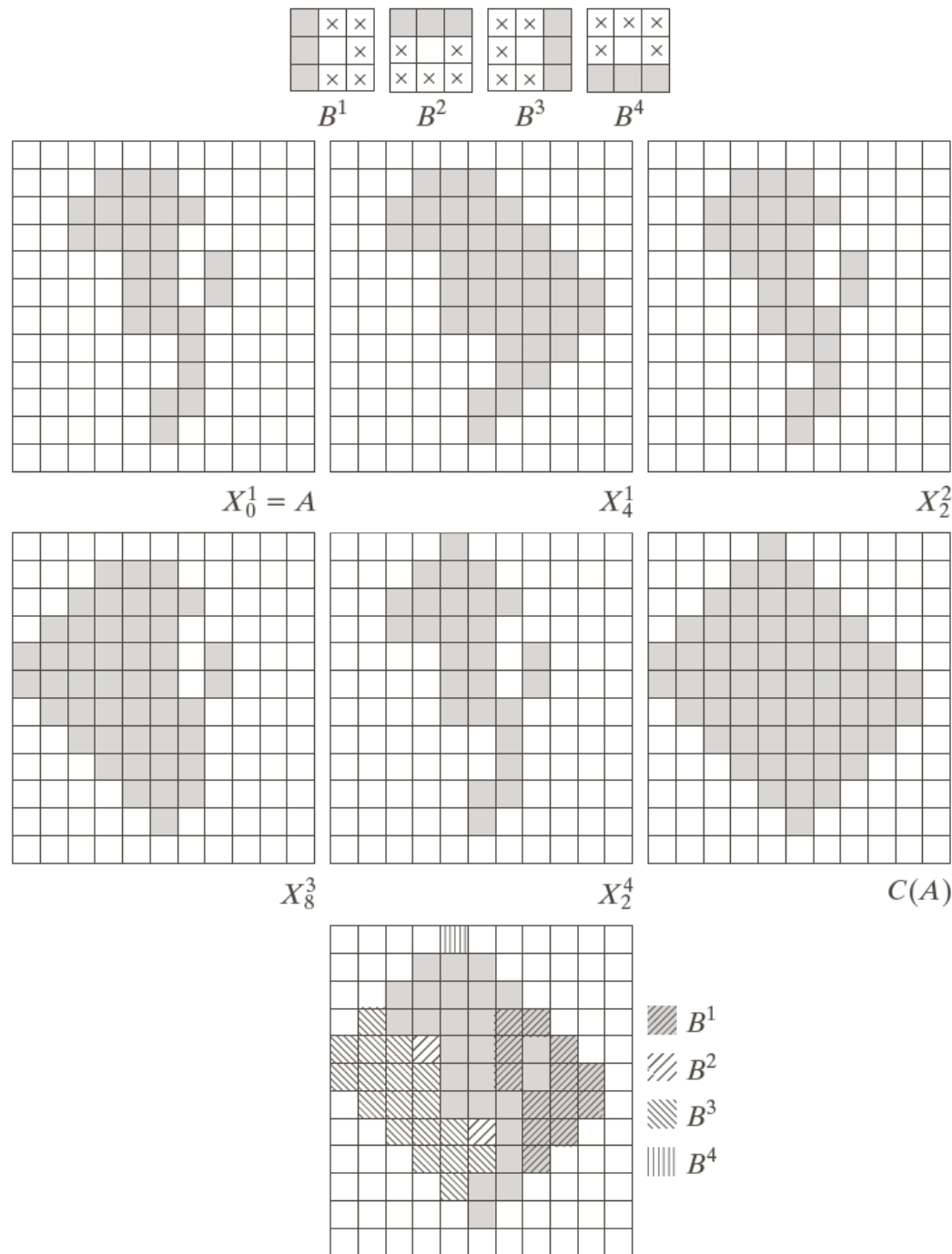
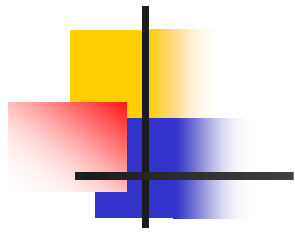
Let B^i , $i = 1, 2, 3, 4$, represent the four structuring elements.
The procedure consists of implementing the equation:

$$X_k^i = (X_{k-1} \circledast B^i) \cup A$$
$$i = 1, 2, 3, 4 \text{ and } k = 1, 2, 3, \dots$$

with $X_0^i = A$.

When the procedure converges, or $X_k^i = X_{k-1}^i$, let $D^i = X_k^i$,
the convex hull of A is

$$C(A) = \bigcup_{i=1}^4 D^i$$



	a	
b	c	d
e	f	g
	h	

FIGURE 9.19 (a) Structuring elements. (b) Set A . (c)–(f) Results of convergence with the structuring elements shown in (a). (g) Convex hull. (h) Convex hull showing the contribution of each structuring element.

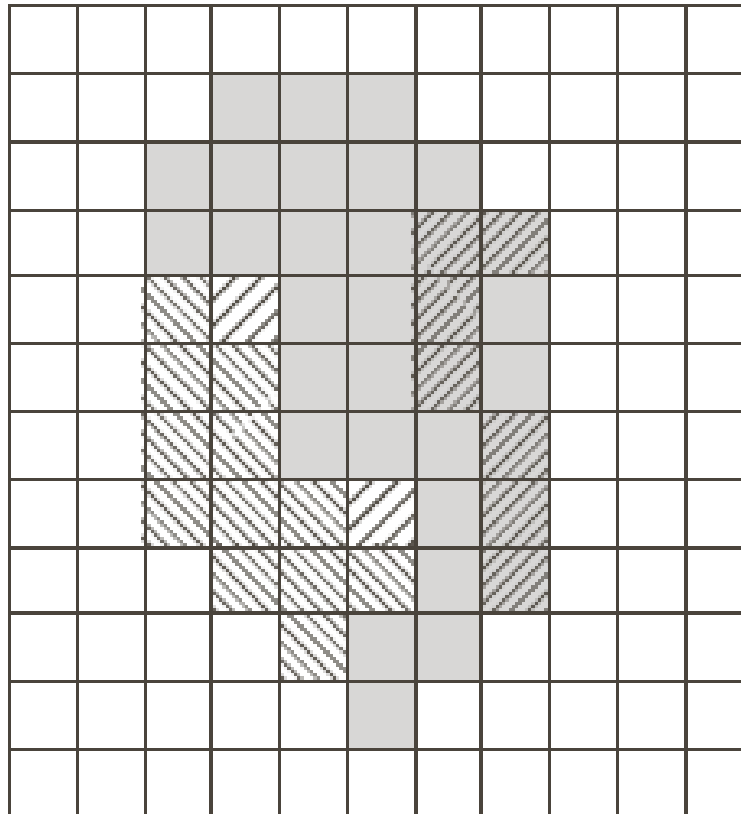
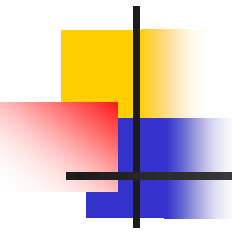


FIGURE 9.20

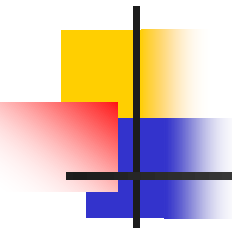
Result of limiting growth of the convex hull algorithm to the maximum dimensions of the original set of points along the vertical and horizontal directions.



Thinning

The thinning of a set A by a structuring element B , defined

$$\begin{aligned} A \otimes B &= A - (A \circledast B) \\ &= A \cap (A \circledast B)^c \end{aligned}$$



Thinning

- A more useful expression for thinning A symmetrically is based on a sequence of structuring elements:

$$\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$$

where B^i is a rotated version of B^{i-1}

The thinning of A by a sequence of structuring element $\{B\}$

$$A \otimes \{B\} = (((...((A \otimes B^1) \otimes B^2)...) \otimes B^n)$$

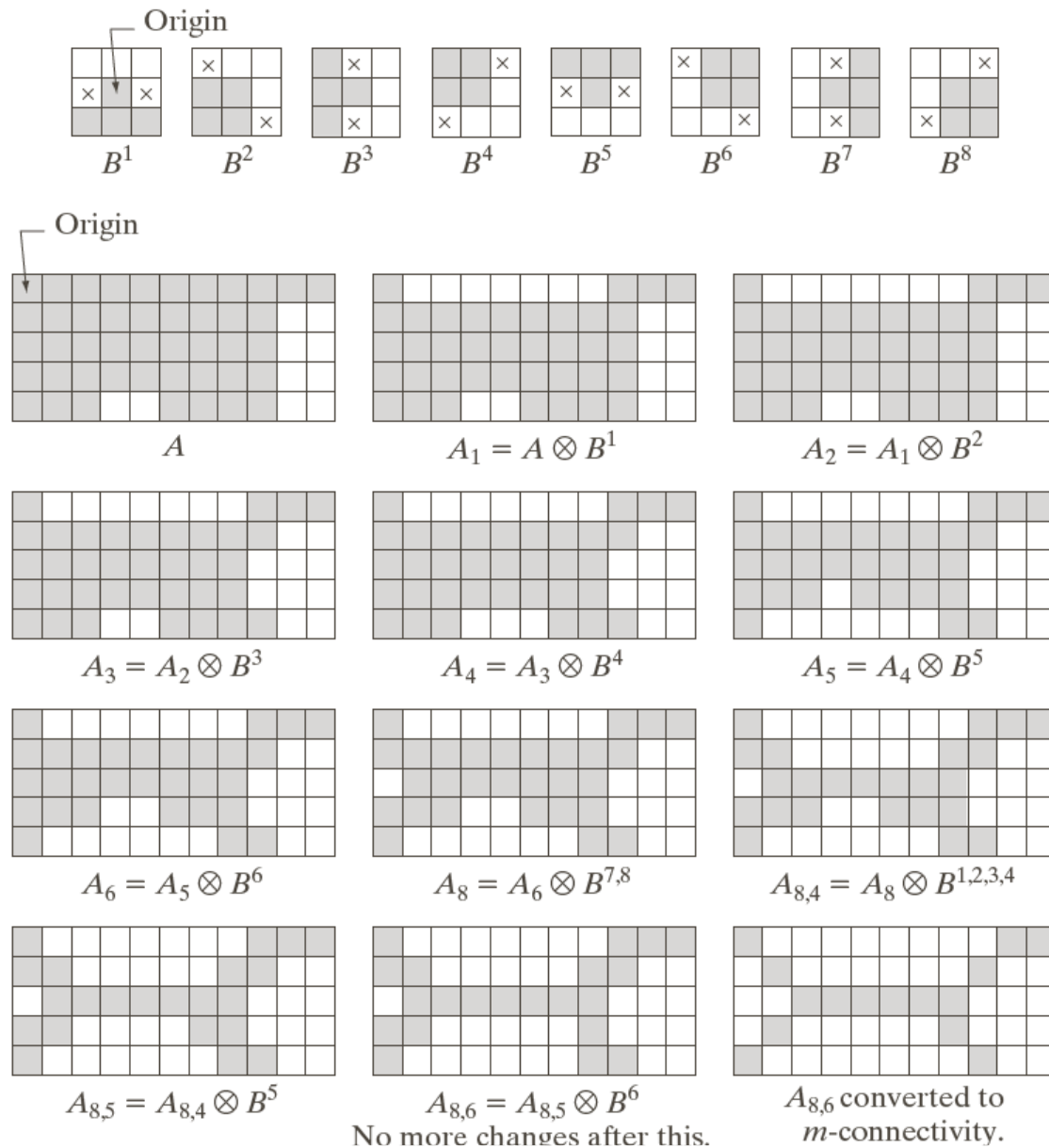
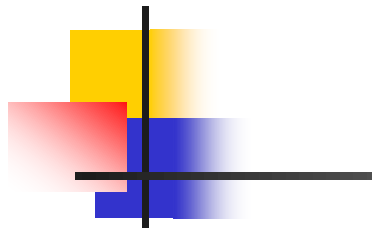
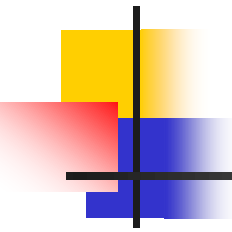


FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A . (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first four elements again. (l) Result after convergence. (m) Conversion to m -connectivity.

a
b c d
e f g
h i j
k l m



Thickening

The thickening is defined by the expression

$$A \odot B = A \cup (A * B)$$

The thickening of A by a sequence of structuring element $\{B\}$

$$A \odot \{B\} = (((\dots (A \odot B^1) \odot B^2) \dots) \odot B^n)$$

In practice, the usual procedure is to thin the background of the set and then complement the result.

Thickening

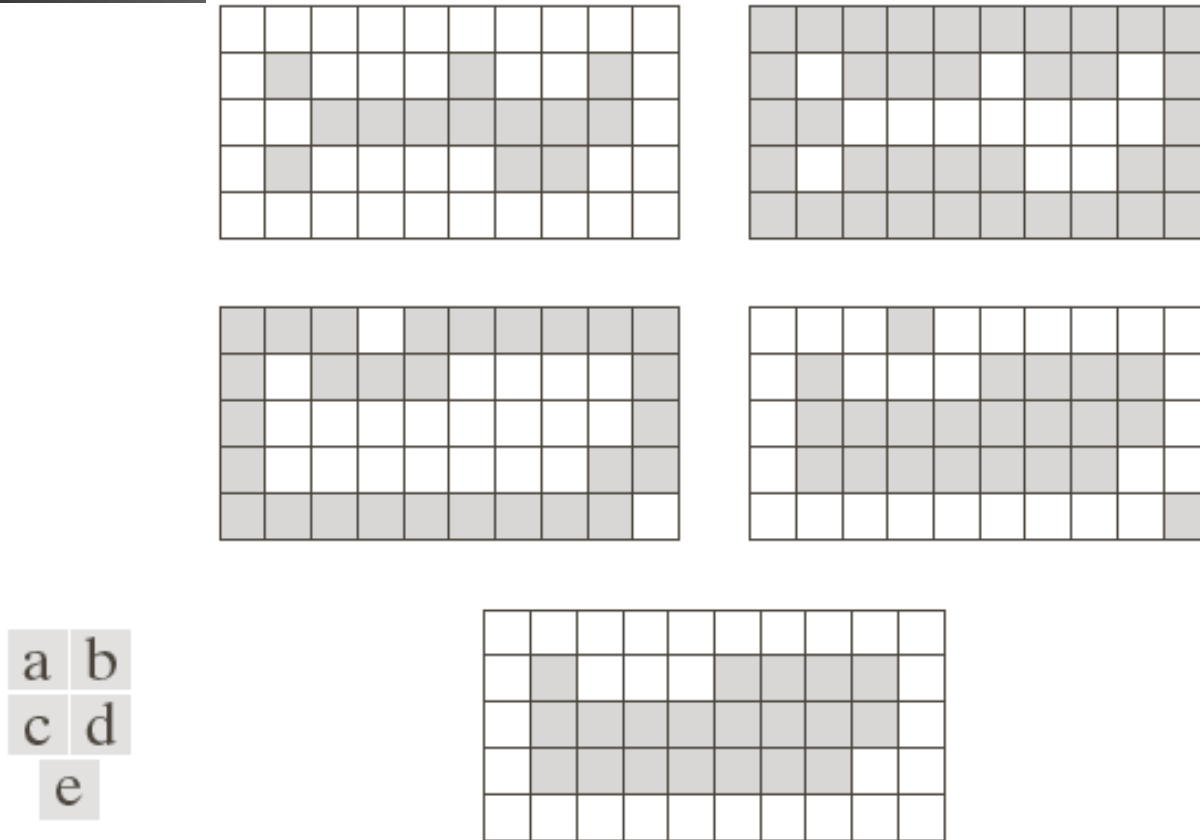
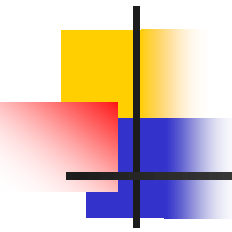


FIGURE 9.22 (a) Set A . (b) Complement of A . (c) Result of thinning the complement of A . (d) Thickened set obtained by complementing (c). (e) Final result, with no disconnected points.



Skeletons

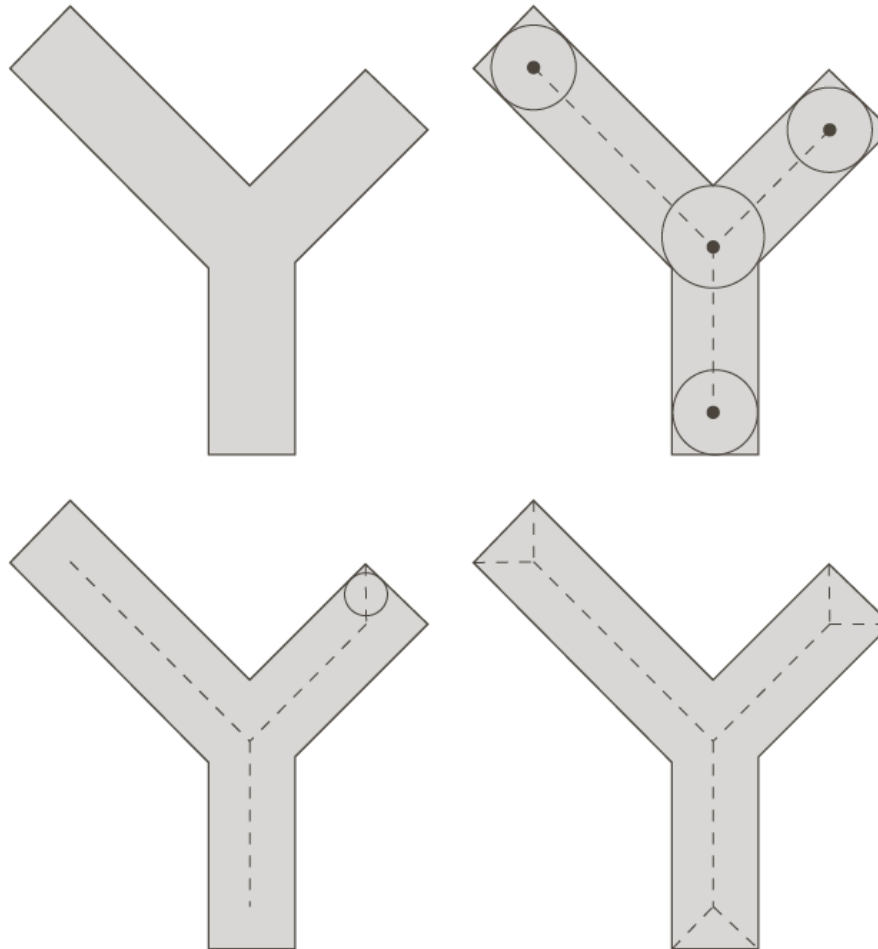
A skeleton, $S(A)$ of a set A has the following properties

- a. if z is a point of $S(A)$ and $(D)_z$ is the largest disk centered at z and contained in A , one cannot find a larger disk containing $(D)_z$ and included in A .

The disk $(D)_z$ is called a maximum disk.

- b. The disk $(D)_z$ touches the boundary of A at two or more different places.

Skeletons



a	b
c	d

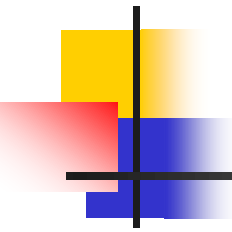
FIGURE 9.23

(a) Set A .

(b) Various positions of maximum disks with centers on the skeleton of A .

(c) Another maximum disk on a different segment of the skeleton of A .

(d) Complete skeleton.



Skeletons

The skeleton of A can be expressed in terms of erosion and openings.

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

with $K = \max\{k \mid A \ominus kB \neq \phi\}$;

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$$

where B is a structuring element, and

$$(A \ominus kB) = ((\dots((A \ominus B) \ominus B) \ominus \dots) \ominus B)$$

k successive erosions of A .

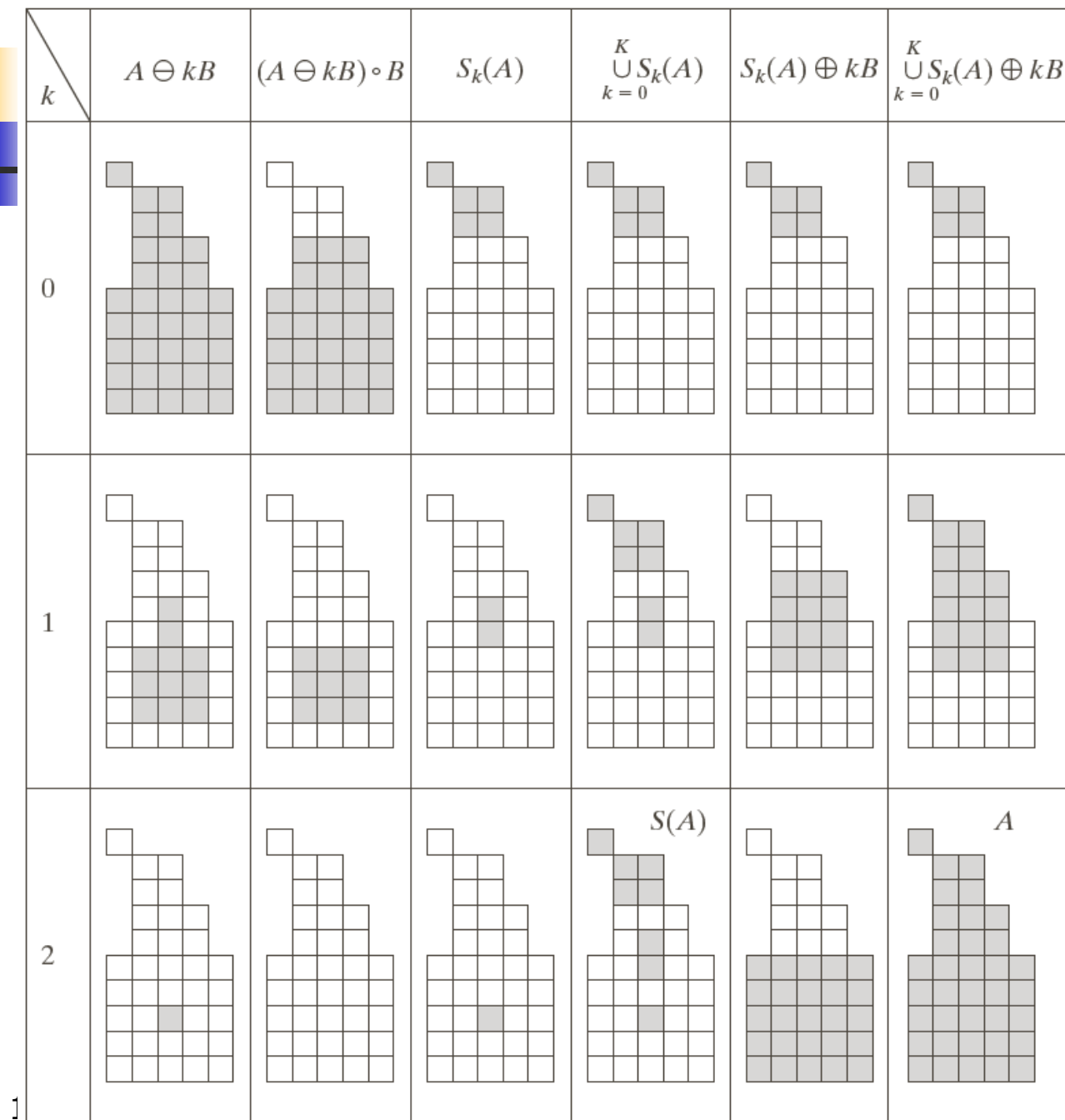
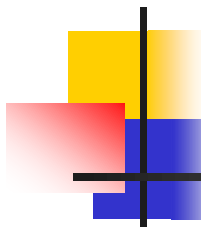
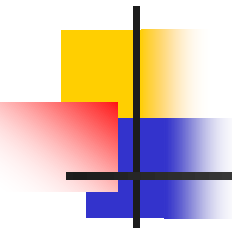


FIGURE 9.24
Implementation of Eqs. (9.5-11) through (9.5-15). The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.



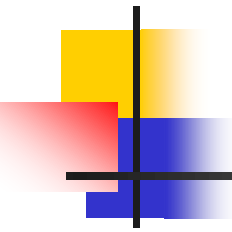
Skeletons

A can be reconstructed from the subsets by using

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

where $S_k(A) \oplus kB$ denotes k successive dilations of A .

$$(S_k(A) \oplus kB) = ((\dots((S_k(A) \oplus B) \oplus B) \dots \oplus B)$$



Thanks
